

Monuments

Registan Square izwi cyane i Samarkand irimo monuments N zihagaze ku murongo unyura muri center y'iyi square. Monuments zanditswe kuva kuri 0 kugeza kuri $N - 1$. Monument i (kuri $0 \leq i < N$) iri kuri integer coordinate $X[i]$. Center ya square iri kuri coordinate 0.

Abubatsi bakeneye symmetry iri perfect ufatiye muri center. Barashaka kwimura zimwe muri monuments (bishoboka ko ari zeru) kugira ngo configuration ya nyuma ibe symmetric ufatiye kuri coordinate 0. Mu bundi buryo:

- Buri monument igomba gushyirwa kuri integer coordinates.
- Buri koordinaten ishobora kuba irimo **zeru, imwe, cyangwa monuments nyinshi**.
- Kuri buri integer $x > 0$, umubare wa monuments ziri kuri coordinate x ugomba kuba ungana n'umubare wa monuments kuri coordinate $-x$. Umubare uwo ari wo wose wa monuments (bishoboka ko ari zeru) ushobora kuboneka kuri coordinate 0.

Ariko, M monuments ni iza cyera cyane (ancient) kandi zoroshye cyane ku buryo zidashobora kwimurwa. Indices zazo ni $P[0], P[1], \dots, P[M - 1]$. Izi M monuments zigomba kuguma kuri original coordinates zazo. Monuments $N - M$ zisigaye zishobora kwimurirwa kuri integer coordinate iyo ari yo yose.

Cost cyo kwimura monument imwe kuva kuri coordinate a ujya kuri coordinate b ni absolute difference $|a - b|$. Total cost ni igiteranyo cya costs za monuments zose zimurwa.

Task yawe ni ugushaka minimum cost yo gukora positions z'ama monuments ziri symmetric ufatiye kuri 0, cyangwa ukamenya ko bidashoboka kugera kuri iyi symmetry hakurikijwe amabwiriza yatanze.

Details Za Implementation

Ugomba ku implementinga procedure ikurikira:

```
long long get_cost(std::vector<int> X, std::vector<int> P)
```

- X : sorted array ifite length N isobanura initial coordinates za monuments.
- P : sorted array ifite length M ifite indices za ancient monuments.
- Iyi procedure iba called rimwe gusa kuri buri test case.

Iyi procedure igomba ku returninga integer imwe: minimum total cost yo gutuma positions ziba symmetric, cyangwa -1 niba bidashoboka.

Constraints

- $1 \leq N \leq 500\,000$
- $0 \leq M \leq N$
- $-10^9 \leq X[0] \leq X[1] \leq \dots \leq X[N-1] \leq 10^9$
- $0 \leq P[0] < P[1] < \dots < P[M-1] < N$

Subtasks

Subtask	Score	Additional Constraints
1	3	$M = N$
2	4	$M = 0$
3	5	$X[P[j]] < 0$ for each j such that $0 \leq j < M$.
4	6	$N \leq 10$
5	5	$N \leq 19$
6	5	$N \leq 32$
7	13	$N \leq 200$
8	17	$N \leq 4000$
9	13	$M \leq 4000$
10	29	No additional constraints.

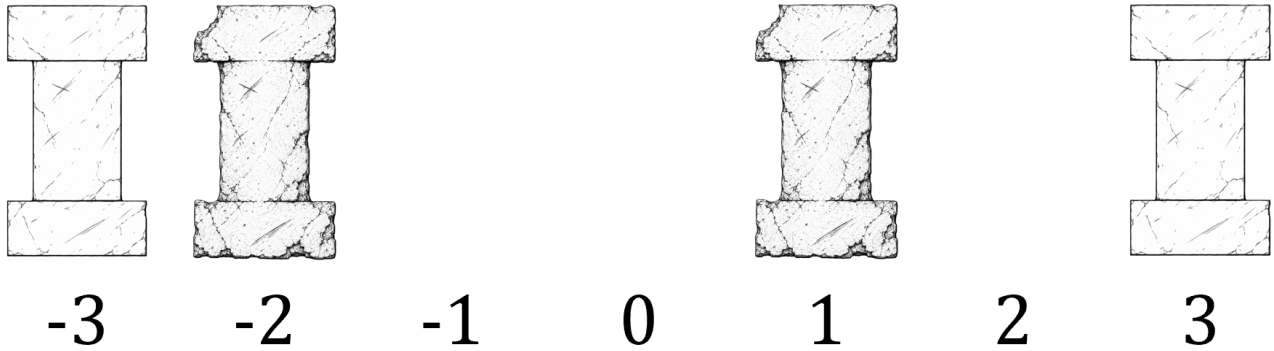
Ingero

Urugero rwa 1

Reka dufate call ikurikira:

```
get_cost([-3, -2, 1, 3], [1, 2])
```

Initial configuration ya monuments igaragaza mu gishushanyo gikurikira.

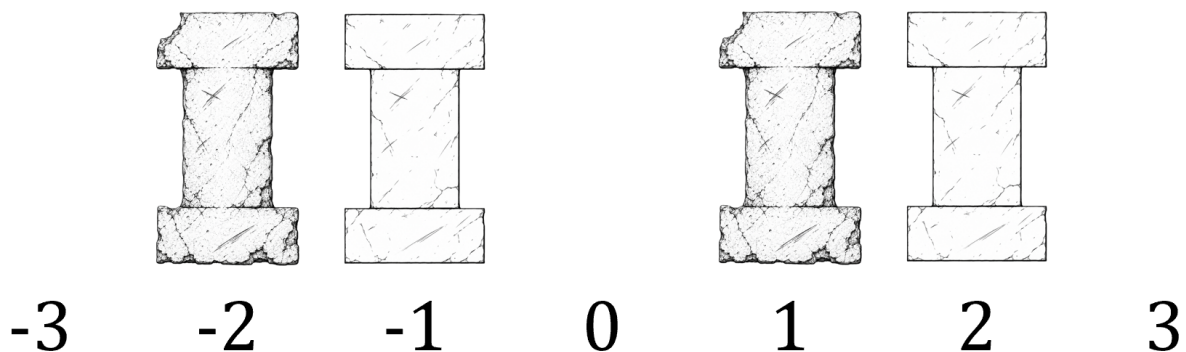


Hari $N = 4$ monuments, initially ziri located kuri coordinates $[-3, -2, 1, 3]$. Indices za ancient monuments ni $P[0] = 1$ na $P[1] = 2$. Bivuze ko, monuments ziri kuri coordinates -2 na 1 zitaba moved. Monuments zisigaye kuri coordinates -3 na 3 zaba moved.

Kugira ngo tugera kuri symmetry dukoresheje minimum total cost, tugomba ku movinga monuments mu buryo bukurikira:

- Movinga monument kuva kuri coordinate -3 kugera kuri -1 , ukoresha cost ingana na $|(-3) - (-1)| = 2$.
- Movinga monument kuva kuri coordinate 3 kugera kuri 2 , ukoresha cost ingana na $|3 - 2| = 1$.

Nyuma y'izi moves, configuration ihita iba symmetric.



Total movement cost ni $2 + 1 = 3$, bityo rero procedure igomba ku returninga 3.

Urugero rwa 2

Reka dufate call ikurikira:

```
get_cost([2, 2, 2, 3], [])
```

Hano, $M = 0$, bivuzeko nta ancient monument n'imwe ihari, kandi monuments zose zaba moved. Igisubizo kimwe gikoreshe minimum total cost ni ukumoving monuments zose kuri coordinate 0. Cost ya muri monument ni:

- $|2 - 0| = 2$ kuri monuments 0, 1, and 2.

- $|3 - 0| = 3$ kuri monument 3.

Total cost ni $2 + 2 + 2 + 3 = 9$. Bityo, procedure igomba ku returninga 9. Umenye ko hari ibindi bisubizo bikoresha total cost ingana n'iyi tubonye.

Urugero rwa 3

Reka dufate call ikurikira:

```
get_cost([1, 2, 3, 4], [0, 1, 2, 3])
```

Monuments 4 zose ni ancient kandi ntago zaba moved. Kuva coordinates zazo zose ziri positive, ntago bishoboka gukora configuration iri symmetric ufatiye kuri 0. Bityo, procedure igomba kureturninga -1 .

Sample Grader

Input Format:

```
N M
X[0] X[1] ... X[N-1]
P[0] P[1] ... P[M-1]
```

Umenye ko umurongo wa gatatu ushobora kuba empty mugihe M ari 0.

Output Format:

```
C
```

Hano, C ni value iba returned na `get_cost`.